



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

Subject: Probability & Statistics

Subject Code: 25SMT-241

Semester: 3rd

Batch: 2025

Assignment -1

Last date of Submission:

Total Marks: 10

***Note: Every student needs to submit an assignment as per the allotted set mentioned in the attached PDF.**

Even UID: - Set1

Odd UID: - Set 2

Set – 1

S. No.	Question	Marks	COs	BT Level																				
1	Prove that the Algebraic sum of deviations of a set of observations from their arithmetic mean is zero.	3 marks	CO4	Level 3																				
2	Calculate the first four moments of the following distribution about the mean and hence find β_1 and β_2 . <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td><td>56</td><td>28</td><td>8</td><td>1</td></tr></table>	x	0	1	2	3	4	5	6	7	8	f	1	8	28	56	70	56	28	8	1	3 marks	CO4	Level 4
x	0	1	2	3	4	5	6	7	8															
f	1	8	28	56	70	56	28	8	1															
3	In a binomial distribution consisting of 5 independent trails, probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find parameter p of the distribution.	4 marks	CO5	Level 4																				



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Set – 2

S. No.	Question	Marks	COs	BT Level
1	Define Moments. Established relationship between moments about mean and moments about arbitrary point.	3 marks	CO4	Level 3
2	Students of a post graduate class were given a general aptitude test and their marks were found to be normally distributed with mean 60 and standard deviation 5. What percentage of students scored? (i) more than 60 marks? (ii) less than 55 marks? (iii) between 45 and 65 marks?	3 marks	CO4	Level 4
3	A manufacture of electric goods packs fuses in the boxes of 10 each and 2000 such boxes were sold. The previous experiences shows that 5% of the fuses are defectives. Using Poisson distribution, find how many boxes will contain (i) no defective fuse, (ii) more than one defective fuses.	4 marks	CO5	Level 4